Introduction to Computer Security

English study summary • Based on the supplied course notes (Chapter 1, 2025–2026)

1. Why Computer Security Matters

The growth of Internet traffic and digital activity brings major benefits, but also increases security risks. Security is not only an IT department responsibility: it helps preserve user trust in the long term, maintain system consistency in the medium term, and ensure that people can access only the information they are entitled to in the short term.

The lecture cites historical figures and incidents to illustrate the scale of cyber risk. These statistics refer to the years stated in the original notes, not current measurements.

- Cyberattacks can cause financial loss, data theft or alteration, service outages, unauthorized system use, and even risks to human life.
- Examples in the notes include WannaCry (2017), NotPetya (2017), the Marriott data breach (2018), attacks targeting domains and organizations (2019), and the Capital One data theft (2019).
- Security measures must include technology, procedures, legal and organizational controls, and user education.

2. Core Security Vocabulary

Term	Meaning	Simple example
Vulnerability	A weakness or security flaw in a system; not every flaw is necessarily exploitable.	An unpatched software flaw.
Attack	A method used to exploit a vulnerability.	Using a flaw to gain unauthorized access.
Countermeasure	A procedure or technique intended to fix a vulnerability or stop a particular attack.	Applying a security patch.
Threat	An adversary capable and motivated to carry out an attack by exploiting a vulnerability.	An attacker targeting a vulnerable server.

3. Six Main Security Objectives

Objective	What it ensures
Authentication	Verifies a user's or entity's identity before communication or access.
Confidentiality	Only authorized parties can understand the data; encryption is commonly used.
Integrity	Data remains accurate and unaltered, whether intentionally or accidentally.
Non-repudiation	A sender or receiver cannot credibly deny having sent or received a transaction.
Availability	Systems, services, and resources remain accessible when needed.
Access control	Access rights are restricted according to each entity's permissions.

4. Attack Categories and Scenarios

Four broad attack categories

- **Interception:** unauthorized observation or copying of information; primarily threatens confidentiality.
- **Interruption:** makes data or a service unavailable; threatens availability.
- **Modification:** changes data, programs, or messages; threatens integrity.
- **Fabrication:** inserts false messages or records; threatens authenticity.

Passive vs. active attacks

- **Passive attacks** observe communications without changing data or system operation. Examples include capturing message contents and traffic analysis. They can be difficult to detect.
- **Active attacks** alter or create data flows. Examples include masquerading as another entity, replaying captured data, modifying or reordering messages, and denial of service.

5. Common Attack Techniques

Technique	Summary
Backdoor	A hidden entry point into a program or system.
Sniffing	Capturing data as it travels across a communication link, especially when it is unencrypted.
Spoofing	Sending packets or messages that appear to come from a trusted or known source.
Bounce attack	Using another computer as an intermediary to conceal the attacker's identity.
Man-in-the-middle	Positioning between two communicating parties to intercept or relay their exchanges.
Denial of service (DoS)	Overwhelming or disabling a service so legitimate users cannot use it.
War-driving	Searching for and attempting to access inadequately protected wireless networks.
Logic bomb	Hidden code that triggers harmful actions when a specified time or event occurs.
Spam and hoaxes	Unsolicited bulk messages; hoaxes try to provoke or persuade people to spread false claims.
Phishing and scams	Impersonating a trusted party or using deceptive stories to steal personal, account, or payment information.

6. Attacker Profiles

- **Black hats:** carry out unauthorized, harmful, or profit-driven attacks.
- **White hats:** test systems to find vulnerabilities and improve security, with a defensive purpose.
- **Grey hats:** combine elements of both approaches; intentions may not be malicious, but actions can still violate rules or laws.
- **Script kiddies:** use publicly available tools or exploits with limited understanding, often to cause disruption or show off.
- **Academic / free-software hackers:** emphasize technical exploration, open-source principles, and sharing knowledge.

The notes also describe different attacker capabilities, from sending forged messages to listening to, disrupting, modifying, or relaying communications.

7. Malware and Related Threats

Threat	Key idea
Virus	Infects files and spreads when infected files are shared or transferred.
Worm	Replicates and spreads across networks without requiring the user to manually pass an infected file.
Adware	Displays advertisements; some forms track browsing. The notes distinguish potentially unwanted adware from clearly malicious software.
Spyware	Secretly gathers information such as keystrokes, browsing habits, or login details.
Ransomware	Encrypts or locks data and demands payment to restore access.
Bots and botnets	Bots automate actions; a botnet is a group of compromised devices controlled remotely, often for spam, data theft, or DDoS attacks.
Rootkit	Hides its presence while enabling privileged or remote control of a system.
Trojan horse	Pretends to be legitimate software or a file but performs harmful actions after installation.
Cryptojacking	Secretly uses a victim's computing resources to mine cryptocurrency.
Bugs / security flaws	Programming errors are not themselves malware, but security bugs can let attackers bypass defenses.

8. Defensive Mechanisms

- **Authentication:** identify users and verify that claimed identities are genuine. Identification states who the user claims to be; authentication checks that claim.
- **Access control:** grants or denies actions on resources according to the security policy and the subject's permissions.
- **Steganography:** hides a secret message inside an ordinary-looking carrier such as an image, audio file, video, or text. Encryption hides the message's meaning; steganography aims to hide the fact that a message exists.
- **Encryption:** transforms plaintext into ciphertext so it is unreadable without the appropriate key. The notes describe symmetric encryption (one shared key) and asymmetric encryption (a public/private key pair).
- **Digital signature:** helps verify the origin and integrity of digital data.
- **Hash function:** maps data of any size to a fixed-size fingerprint; changing the input should change the resulting hash.
- **Digital certificate:** acts like a digital identity credential and can support authentication, access control, and electronic signatures.
- **Antivirus:** detects, blocks, or removes malware, but does not stop every misuse by authorized users or legitimate software.
- **Firewall:** filters network traffic between external and local networks, but cannot block every attack, especially one that uses an authorized connection or bypasses the firewall.
- **Intrusion Detection System (IDS):** detects suspicious or abnormal activity on a host or network.
- **Logging:** records events in chronological order to support monitoring and later investigation.

9. Main Takeaway

A system is only as secure as its weakest link. Security must therefore be addressed together at four levels:

- **People:** users need awareness and training.
- **Technology:** systems and software should be secure and patched.
- **Data and permissions:** users should receive only the access rights they need.
- **Physical environment:** facilities and hardware must be protected, not just software and networks.

Study cue: be able to define the four core terms (vulnerability, attack, countermeasure, threat), name the six security objectives, distinguish passive from active attacks, match attack categories to the security property they threaten, and explain the purpose and limits of each defensive mechanism.